Calculus II

Tutorial #7 31 Mar – 04 Apr 2025

Topics:

- Basic knowledge of differential equations
- Solutions to differential equations

• **True-False Review**. Decide if the given statement is **True** or **False**, and justify your answer.

1. If room temperature is 70°F, then an object whose temperature is 100°F at a particular time cools faster at that time than an object whose temperature at that time is 90°F.
2. According to Newton's law of cooling, the temperature of an object eventually becomes the same as the temperature of the surrounding medium.
3. A hot cup of coffee that is put into a cold room cools more in the first hour than the second hour.

• **Problems**.

4. Verify that
$$y(t) = C_1 \cos \omega t + C_2 \sin \omega t$$
is a solution to the differential equation
$$\frac{d^2y}{dt^2} + \omega^2 y = 0.$$
(this is a differential equation for some motion). Write the solution in a form $y(t) = A\cos(\omega t + \phi)$, where A is some constant (which is called the **amplitude** of the motion).
5. Show that the given relation defines an implicit solution of the differential equation, where C is an arbitrary constant:
$$e^{xy} - x = C,\ y' = \frac{1 - ye^{xy}}{xe^{xy}}.$$
6. Verify that the relation $x^2 + y^2 - 4 = 0$ defines an implicit solution of the nonlinear differential equation
$$\frac{dy}{dx} = -\frac{x}{y}.$$

7. Find the general solution of the following differential equations and the maximum interval on which the solution is valid:

$$y'' = x^n,$$

where $n \in \mathbb{Z} = \{0, \pm 1, \pm 2, \ldots\}$.

8. Show that $y = \dfrac{1}{x}\displaystyle\int_1^x \frac{e^t}{t}\,dt$ is a solution of the differential equation

$$x^2y' + xy = e^x.$$

9. Let a be a nonzero real number.

   (a) Verify that if b is an arbitrary constant, then

   $$y = (x-b)^a \tag{1}$$

   is a solution of

   $$y' = ay^{(a-1)/a} \tag{2}$$

   on (b, ∞).

   (b) Suppose that $a < 0$ or $a > 1$. Can you think of a solution of (2) which is <u>not</u> of the form (1)?

10. (a) Verify that if b is any real number, then

    $$y = b^2 + bx + 2b + 1 \tag{3}$$

    satisfies

    $$y' = \frac{-(x+2) + \sqrt{x^2 + 4x + 4y}}{2} \tag{4}$$

    on some open interval. Identify the open interval.

    (b) Verify that

    $$y_1 = \frac{-x(x+4)}{4}$$

    also satisfies (4) on some open interval, and identify the open interval. (Note that y_1 <u>cannot</u> be obtained by selecting a value of b in (3)).

END

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Answers to selected problems

1. True
2. False

3. True

7. (a) If $n = -1:\ y(x) = x \ln|x| + Cx + D$

   (b) If $n = -2:\ y(x) = -\ln|x| + Cx + D$

   (c) If $n \neq -1, -2:\ y(x) = \dfrac{x^{n+2}}{(n+1)(n+2)} + Cx + D,$

   where C, D are arbitrary constants.

9. (b) If $a > 1$ or $a < 0$, then $y \equiv 0$ is a solution of (2) on $(-\infty, \infty)$.